(Please write your Enrollment Number) Enrollment No._____

MID-TERM EXAMINATION
(B.Tech CSE/ECE/ ECE-AI) 1st semester
(October, 2023) OFF LINE mode

Subject Code: BCS 110	Subject: Programming in C Language
Time : 1 ½Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

Q1 (2.5*4)

(a) What are the differences between the Assembler and interpreter?

(b) Convert 8TB into Mega Bytes.

(c) Explain ternary operator with example.

(d) Find the output of the following program

```
#include <stdio.h>
int main()
{
   int i = -1, j = 0, k = 3, m;
   m = i++ && ++j || k++;
   printf("%d\t%d\t%d\t%d",m,i,k,j);
   return 0;
}
```

Q2 (Attempt any Two Parts) UNIT-1 (5,5)

(a) Write the output of following two programs. Show necessary execution steps.

i.
```
#include <stdio.h>
 int main()
 {
 int a=5,b=5;
 printf ("%d,%d\t",++a,b--);
 printf ("%d,%d\t",a,b);
 printf ("%d,%d\t", ++a,b++);
 printf ("%d, %d\n",a, b);
 }
```

ii.
```
#include <stdio.h>
 int main()
 {
 int a=15,b=13,c=16,x,y;
 x=a-3%2+c*2/4%2+b/4;
 y=a=b+5-b+9/3;
 printf ("x = %d, y = %d\n", x, y) ;
 }
```

(b) Accept any two numbers, if the first number is greater than second then print the sum of these two numbers, otherwise print their difference. Write this program using ternary operator.

(c) Explain the following Primary and Secondary memory.

Q3 (Attempt any Two Parts) UNIT-2 (5,5)

(a) Write the flow chart of break and continue statements.

(b) Write the output of following program:

i) a = 8, b = 5, c = 8, d = 3, e = 65, f = 10, g = 2, h = 5, k = 2

```
a - b + c / d == e / f - g + h % k
```

ii) a = 8, b = 3, c = 2, d = 3, e = 2, f = 11

```
a - b || ( a- b * c ) + d && e - f % 3
```

(c) Write a C program to design calculator with basic operations using switch.